Module Six Assignment

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CS-370: Current and Emerging Trends in Computer Science

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The main objective of the cartpole problem is to use a moving cart to keep a pole in a vertical position. For this assignment, the cartpole problem was solved using a reinforcement learning algorithm called A2C. This algorithm works by allowing the AI to learn a policy (which is a strategy that tells the AI what actions it should take) using the actor network directly while also using a critic network to estimate the value of a given move and either reward or penalize that move.

The cartpole problem can also be solved using the REINFORCE algorithm or the Q-learning algorithm. The REINFORCE algorithm only uses policy-based learning to teach the AI how to accomplish its objective, whereas the Q-learning algorithm uses the value-based approach. Both types of reinforcement learning have advantages and disadvantages:

Value-Based Learning:

- Model-free: doesn't require a pre-existing model to learn how to complete a task
- Sample efficiency: the AI can learn from a limited number of previous experiences
- Slow learning speed and high computational costs can make them inefficient at solving complex problems.

Policy-Based Learning:

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- Optimized Strategy: the direct use of a policy can lead the AI to become more effective and intuitive in many scenarios.
- Improved convergence: allows for more stability during the reinforcement learning process.
- Due to the need for large amounts of data, policy-based learning might not be as sample-efficient.

References

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